SAI TEJA ADUSUMILLI

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Summary: Data Analytics professional with experience in Python, ML, NLP, and SQL, building data models, ETL pipelines, and dashboards, with a focus on applying analytical methods to help businesses improve performance and make better decisions.

EDUCATION

California State University, Sacramento — Sacramento, CA
Master of Science in Business Analytics, GPA: 3.9/4.0 — Aug 2024 – Dec 2025
Relevant Coursework: *Data Analytics, Data Visualization, Data Mining, Machine Learning, Data Management, Big Data for Business*

Krishna University — Vijayawada, India
Master of Science in Physics, GPA: 4.0/4.0 — July 2017 – April 2019
Bachelor of Science in Mathematics, Physics, and Computer Science, GPA: 3.99/4.0 — July 2014 – April 2017

SKILLS

- **Languages:** Python (scikit-learn, PyTorch, NumPy, Pandas), R, SQL, HTML, CSS
- **Machine Learning Techniques:** Decision Trees, Regression, Ensemble Methods, Time Series, NLP (TF-IDF, BERTopic)
- **Data Visualization Tools & Applications:** Tableau, Power BI, Matplotlib, Seaborn, Plotly, Streamlit
- **Technologies & Tools:** AWS (S3, SageMaker, Rekognition), Git, VS Code, Microsoft Office Suite (Excel, PowerPoint, Word)
- **Databases & Data Engineering:** MySQL, PostgreSQL, Firebase, ETL/ELT pipelines
- **Business Skills:** Financial Accounting, Risk Management, Banking Regulations, Customer Relationship Management (CRM)

EXPERIENCE

California State University, Sacramento — Sacramento, CA
Volunteer Research Assistant - Gen AI Project — Aug 2024 – Jan 2025

- Reduced 2,000 research articles to a focused set of 1,300 using BERTopic, improving topic coherence and clustering quality.
- Developed a Python pipeline using TF-IDF and rule-based NLP to classify articles under the STS framework for semantic categorization supporting AI-driven analytics.

State Bank of India — Bangalore, India
Associate — June 2022 – Jan 2023

- Led a Risk-Focused Internal Audit (RFIA) across loan and branch operations, analyzing 8 years of records to assess regulatory adherence, identify control gaps, and mitigate compliance risks; achieved an 'A' rating with zero audit findings.
- Evaluated 60+ loan applications worth $1 million and monitored post-disbursal performance using credit risk frameworks; improved approval efficiency by 35% and reduced defaults through early-risk detection.

Junior Associate — Jan 2021 – May 2022

- Analyzed high-value customer transaction patterns for targeted marketing, boosting investment and lending product sales by 35%.
- Spearheaded dormant portfolio reactivation by coordinating KYC verification for 200+ high and medium risk profiles, ensuring compliance and reducing documentation errors by 30%.

PROJECTS

Credit Risk Insights Dashboard | *Python (LightGBM, XGBoost, CatBoost, Random Forest), Streamlit*

- Developed a credit-risk prediction model by engineering features from **27M+** records and evaluating multiple ensemble algorithms achieving **0.78** AUC-ROC and **69%** recall. The final LightGBM model generates applicant-level probability-of-default scores used for risk stratification and lending decisions.
- Deployed the model into a real-time **Streamlit** dashboard with automated PD scoring, explainable risk indicators (SHAP), and threshold simulation, enabling credit analysts to optimize approval policies and balance business growth against portfolio default risk.

Database & Data Warehouse Development for a Property Management Company | *SQL, ERDPlus, Tableau*

- Developed a MySQL database and data warehouse, implementing ETL processes through advanced SQL queries and stored procedures to manage leasing and tenant data, enabling efficient business reporting.
- Built business intelligence reports using **Tableau**, delivering actionable insights to optimize property management strategies.

SBA Loan Default Prediction & Profit Optimization | *Python (Decision Trees, Ensemble Methods, Regression, Neural Networks)*

- Built machine learning models on **890K** SBA loan records to analyze risk exposure using a cost-sensitive profit framework (+5% repaid, -25% default) with hyperparameter tuning across **14** algorithms to predict defaults and improve profitability.
- Identified Bagging as the top-performing model, estimating $2.64B in net profit with a 0.4 default probability cutoff and approval of 76% lowest risk applicants to maximize lender returns.

Fake News Analysis using NLP | *NLP (n-gram, Sentiment and Stylometric Analyses, Topic Modeling (LDA))*

- Conducted a comparative NLP analysis of 12 YouTube transcripts using a 5-method pipeline to capture linguistic features across sentiment, structure, and readability, distinguishing misinformation from authentic journalism.
- Achieved statistically significant results ($p<0.05$) as fake news showed **26%** less vocabulary complexity and the use of fear-based language while real news used balanced scientific terminology.